Answers learning task 14

2026 – 2027

Exercise 8.29: A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 30$ mL and a standard deviation of $s = 0.5$ mL. Find a $95\,\%$ CI on the mean volume of syrup dispensed.

Answer

Whenever we want a two-sided $100 \cdot (1 - \alpha)$ % confidence interval for the mean μ (in case σ is unknown), we use the formula

$$[\overline{x} - t_{\alpha/2;n-1} \cdot \frac{s}{\sqrt{n}}, \overline{x} + t_{\alpha/2;n-1} \cdot \frac{s}{\sqrt{n}}]$$

Notice that we also do not have access to the value of σ, so we also have to estimate σ with the sample standard deviation s. Here, if we want a $95\,\%$ confidence interval, that means $100 \cdot (1 - \alpha) = 95$, or $\alpha = 0.05$. The corresponding value of $t_{\alpha/2;n-1}$ can be found by taking the $1 - \alpha/2$-percentile of the t-distribution with df $= n - 1 = 24$ degrees of freedom:

$$t_{\alpha/2;n-1} = t_{0.025;24} = \text{InvT}(\text{area} = 1 - \alpha/2 = 0.975, \text{df} = 24) \approx 2.0639.$$

Filling in $\overline{x} = 30$, $s = 0.5$, $n = 25$ and $\alpha = 0.05$ gives $[29.7936, 30.2064]$.

Exercise 8.37: The compressive strength of concrete is being tested by a civil engineer who tests 10 specimens and obtains the following data:

2216	2237	2249	2204
2225	2301	2281	2263
2318	2255		

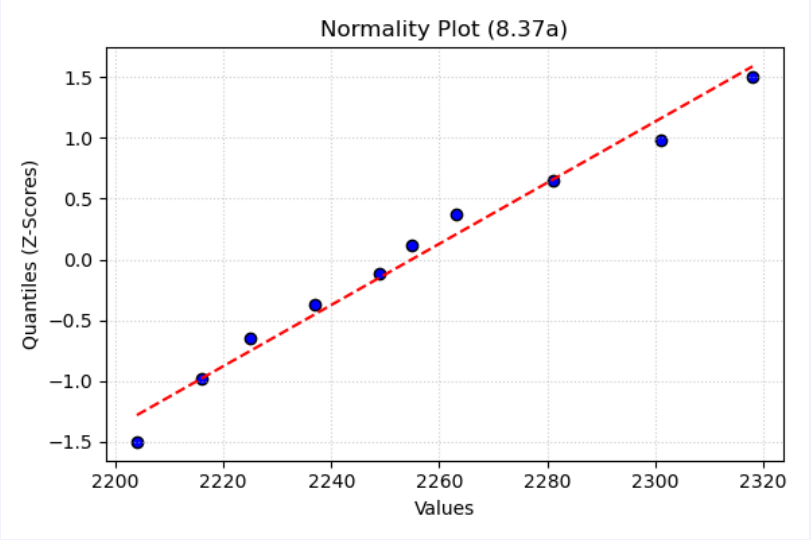
(a) Check the assumption that compressive strength is normally distributed. Include a graphical display in your answer.

(b) Construct a $95\,\%$ two-sided confidence interval on the mean strength.

(c) Construct a $95\,\%$ lower confidence bound on the mean strength. Compare this bound with the lower bound of the two-sided confidence interval and discuss why they are different.

Answer

(a) Normal probability plot:



The points lie reasonably close to a straight line, indicating that the compressive strength data can be assumed to be normally distributed.

(b) First, we need to compute the sample mean $\overline{x}$ and the sample standard deviation s:

$$\overline{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{2216 + 2237 + \ldots + 2255}{10} \approx 2254.9000.$$

To compute the sample standard deviation s, we first compute the sample variance s^2:

$$\begin{aligned} s^2 &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2 + \ldots + (x_n - \overline{x})^2}{n-1} \\ &= \frac{(2216 - 2254.9000)^2 + (2237 - 2254.9000)^2 + \ldots + (2255 - 2254.9000)^2}{10-1} \\ &\approx 1356.3222. \end{aligned}$$

The sample standard deviation s is obtained by taking the square root of the sample variance s^2:

$$s = \sqrt{s^2} = \sqrt{1356.3222} \approx 36.8283.$$

Filling in $\overline{x} = 2254.9$, $s = 36.8283$, $n = 10$ and $\alpha = 0.05$ gives $[2228.5546, 2281.2454]$.

(c) For a lower confidence bound on the mean strength, we can compute a different type of interval. This time it is not symmetric around the sample mean $\overline{x}$, but using the following formula:

$$[\overline{x} - t_{\alpha;n-1} \cdot \frac{s}{\sqrt{n}}, \infty)$$

Notice the difference in the subscript for the t-value: we now use the α-th percentile instead of the $\alpha/2$-th percentile. Filling in $\overline{x} = 2254.9$, $s = 36.8283$, $n = 10$ and $\alpha = 0.05$ gives $[2233.5513, \infty)$.

Exercise 8.39: An article in *Computers & Electrical Engineering* ["Parallel Simulation of Cellular Neural Networks" (1996, Vol. 22, pp. 61–84)] considered the speedup of cellular neural

networks (CNN) for a parallel general-purpose computing architecture based on six transputers in different areas. The data follow:

3.775302	3.350679	4.217981	4.030324	4.639692
4.139665	4.395575	4.824257	4.268119	4.584193
4.930027	4.315973	4.600101		

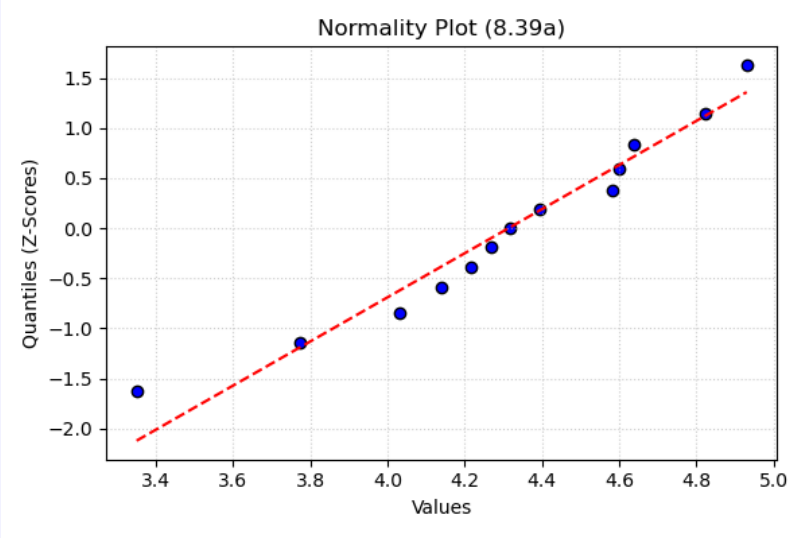
(a) Is there evidence to support the assumption that speedup of CNN is normally distributed? Include a graphical display in your answer.

(b) Construct a $95\,\%$ two-sided confidence interval on the mean speedup.

(c) Construct a $95\,\%$ lower confidence bound on the mean speedup.

Answer

(a) Normal probability plot:



The points approximate a straight line, showing that the speedup is normally distributed.

(b) First, we need to compute the sample mean $\overline{x}$ and the sample standard deviation s:

$$\overline{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{3.775302 + 3.350679 + \ldots + 4.600101}{13} \approx 4.3132.$$

To compute the sample standard deviation s, we first compute the sample variance s^2:

$$\begin{aligned} s^2 &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2 + \ldots + (x_n - \overline{x})^2}{n-1} \\ &= \frac{(3.775302 - 4.3132)^2 + (3.350679 - 4.3132)^2 + \ldots + (4.600101 - 4.3132)^2}{13 - 1} \\ &\approx 0.1873. \end{aligned}$$

The sample standard deviation s is obtained by taking the square root of the sample variance s^2:

$$s = \sqrt{s^2} = \sqrt{0.1873} \approx 0.4328.$$

Filling in $\overline{x} = 4.3132$, $s = 0.4328$, $n = 13$ and $\alpha = 0.05$ gives $[4.0517, 4.5748]$.

(c) For a lower confidence bound on the mean strength, we can compute a different type of interval. This time it is not symmetric around the sample mean $\overline{x}$, but using the following formula:

$$[\overline{x} - t_{\alpha;n-1} \cdot \frac{s}{\sqrt{n}}, \infty)$$

Notice the difference in the subscript for the t-value: we now use the α-th percentile instead of the $\alpha/2$-th percentile. Filling in $\overline{x} = 4.3132$, $s = 0.4328$, $n = 10$ and $\alpha = 0.05$ gives $[4.0993, \infty)$

Exercise 8.41: An article in *Nuclear Engineering International* (February 1988, p. 33) describes several characteristics of fuel rods used in a reactor owned by an electric utility in Norway. Measurements on the percentage of enrichment of 12 rods were reported as follows:

2.94	3.00	2.90	2.75	3.00	2.95
2.90	2.75	2.95	2.82	2.81	3.05

(a) Use a normal probability plot to check the normality assumption.

(b) Find a $99\,\%$ two-sided confidence interval on the mean percentage of enrichment. Are you comfortable with the statement that the mean percentage of enrichment is $2.95\,\%$? Why?

Answer

(a) The data appear to be normally distributed.

(b) First, we need to compute the sample mean $\overline{x}$ and the sample standard deviation s:

$$\overline{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{2.94 + 3.0 + \ldots + 3.05}{12} \approx 2.9017.$$

To compute the sample standard deviation s, we first compute the sample variance s^2:

$$\begin{aligned} s^2 &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2 + \ldots + (x_n - \overline{x})^2}{n-1} \\ &= \frac{(2.94 - 2.9017)^2 + (3.0 - 2.9017)^2 + \ldots + (3.05 - 2.9017)^2}{12 - 1} \\ &\approx 0.0099. \end{aligned}$$

The sample standard deviation s is obtained by taking the square root of the sample variance s^2:

$$s = \sqrt{s^2} = \sqrt{0.0099} \approx 0.0993.$$

Filling in $\overline{x} = 2.9017$, $s = 0.0993$, $n = 12$ and $\alpha = 0.01$ gives $[2.8126, 2.9907]$.

We are comfortable with the statement that the mean percentage of enrichment is $2.95\,\%$, as this value lies inside the confidence interval.